Recall: where the project stands

A memory store for people who cannot keep one, with a camera that does the filing and a second interface for the family member who knows what matters. Version 1 of this report was written while the database was still a schema file. Since then the database is live, the two interfaces talk to each other, the database patch is applied, and a full test pass of about a hundred and forty checks found thirteen defects, every one of them fixed and live. This version says what is proven, what is still only written, what is left for whom, and where the remaining hours of this sprint should go.

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2 interfaces live, and they now share one household	**25 / 25** security checks pass, and zero accessibility violations	**13** defects found by testing today, all thirteen fixed and live	**0** blockers left, the database patch is applied and verified	**1,450** paying households is break even, on the numbers we present

01

What Recall is, in one paragraph

Older people lose the thread of ordinary things: a face without a name, whether the pill was taken, what the hospital letter actually said, where a thing is kept. Every existing answer is a notes app or an alarm clock, and neither survives a week, because the person who forgets cannot file documents. Recall solves the intake problem instead of the storage problem. The keeper points a camera at a letter **because they cannot read the small print**; Recall magnifies it, reads it out loud, finds the date, and filing the card is what happens on the way. A magnifier is a free tool. A magnifier that files is a product.

After the pitch, one thing changed: the person who forgets is not the person who knows what matters. So the family gets an interface of its own, over the same store.

02

Scope and the rules we agreed

THREE DECISIONS THAT FIXED THE SHAPE OF THE PROJECT

DECISION	WHAT IT MEANS
The professor, 7 September	Only the memory module, with the camera folded into it as intake. The mobility module, memory walks, family messaging integrations and face recognition are parked on a roadmap for the end presentation, together with the security and identity work (itsme, FranceConnect and the EU Digital Identity Wallet).
The pitch jury, 8 September	A caretaker interface is needed as well: someone has to add what is not on paper, follow up on what was heard and done, and set the app up in the first place. A record therefore belongs to a **household**, not to a person.

The team, 8 September	Thirteen working hours are left, so: a dev freeze on Thursday at three in the afternoon, a cut list agreed in advance rather than in the panic (web push, export, unlinking a phone, the torch, spoken questions), one rehearsal with a timer, and all testing done by one person so that nobody marks their own homework.

03

What works today, and how we know

"Proven" means somebody ran it against the live database or on a phone and watched it happen. "Written" means the code exists and was reviewed but nothing has exercised it yet. The difference matters: version 1 of this report had nine rows of written and no database at all.

KEEPER APP, INDEX.HTML, THE PHONE THE OLDER PERSON HOLDS

CAPABILITY	STATUS	EVIDENCE OR GAP
Today screen, reminders first, read out loud	PROVEN	Reminders render large, each with read out loud and mark done
All cards, with search, and one card in detail	PROVEN	Six fields, the reminder, the last time it was done, the medical line
Camera with a magnifier up to four times	PROVEN	Runs on a phone. The screen now sizes itself to the device, so nothing has to be scrolled while she is holding a letter up to it
Reads only what the zoom is showing	PROVEN	Fixed today. It used to read the whole frame, which is not what she is looking at
On-device OCR, date parsing, national number stripped	PARTLY	Runs, and the redaction is in code. Accuracy on real folded paper is Luke's test and is not signed off yet
Works with no account, no internet, nothing leaving the phone	PROVEN	With the config empty, no database code is downloaded at all
Installs on the home screen, and now offers to	PROVEN	The offer appears at the one moment somebody competent is holding the phone, which is right after the family links it. iPhone gets the instructions instead, because Safari gives no prompt
Linking the phone: six letters, or a square to look at	PROVEN	The square carries the code in a link, so the phone lands linked with nothing typed
An optional code, four or six numbers, before Recall opens	PROVEN	Set by family on her phone. Never comes on by itself, and a fresh phone code from the family app always unlocks it again, so she cannot be shut out of her own memory
Face ID or a fingerprint instead of the numbers	PARTLY	Built on the phone's own authenticator, offered only where the phone has one. Needs a phone with a face on it to sign off
Family cards arriving on the phone, and binned cards leaving it	PROVEN	Both were broken this morning and both are fixed. Luke re-tests on his own phone

FAMILY APP, HELPER.HTML, THE PHONE THE DAUGHTER HOLDS

CAPABILITY	STATUS	EVIDENCE OR GAP
Sign in with an emailed link, no password	PROVEN	Luke signed in on his own account today

Create a household and pick between households	PROVEN	One press now makes one household. It used to make three
Cards: list, add by typing, photo, the spoken sentence	PROVEN	Luke added a card and it reached the keeper phone
Reminders with repeats, and follow-up	PROVEN	Three values and no more: coming, done, missed
Thirty day bin with restore	PROVEN	There is no delete policy in the database at all, so nothing can be destroyed outright
Who has access, by name, on the keeper side	PROVEN	Names, not a count, so the keeper can recognise them
Move Recall to another phone, in three steps	PROVEN	Tested end to end against the live database, including a replacement phone and removing the old one. Nothing is copied, because cards belong to the household
See which phones are linked, and remove one	PROVEN	Her phones are a separate list from the people, with the date each was linked. Removing one lands in the log she can read
Invite a second family member	WRITTEN	Token that expires after a week. Nobody has used one yet
Activity log the keeper can read	WRITTEN	Insert only for members, no update and no delete policy
Spoken consent screen and the consent record	NOT BUILT	Wednesday morning. Until it exists the upload path has a lawful basis on paper only

The one property worth stating plainly

Isolation is not a promise in this report, it is a test. `tests/rls_test.py` opens two separate sessions, builds a household with a card and a photo in one, and then tries everything the other one could try: read the cards, read the households, read the log, write a row, download the photo, claim a made up code. It also carries the bug testing found on Tuesday: a family member who claims a link code must still be a helper afterwards. Twenty five checks, twenty five green, run against the live project.

04

What testing found today

Testing paid for itself twice over. Luke took the phones: camera, zoom and a real letter on real paper, which is the one thing a browser on a laptop cannot honestly prove. The rest of the plan ran in the browser and against the live database, about a hundred and forty checks, written up in `docs/test-plan.md` with what was actually seen for each one rather than a tick. Thirteen defects, all thirteen fixed the same day. The six below came out of Luke's morning. Seven more came out of the full pass, and two of those broke promises this app makes out loud: it never showed the text it had just read off a letter, and a photo taken on her phone never reached the family at all.

THE SIX FROM THE MORNING, ALL FIXED THE SAME AFTERNOON

#	WHAT WAS SEEN	WHAT WAS ACTUALLY WRONG
1	A card added by the family never arrived on the keeper phone	Three faults in one: the phone pushed before it pulled, the example cards collided with a real one on the way up, and after a successful sync nothing redrew the screen. The error was being swallowed, so the phone looked fine
2	One household appeared three times	The create button could be pressed again while the first request was still in flight
3	A family member silently became the keeper	Claiming a link code in a browser already signed in as family overwrote the role. Now the app refuses, and the database will refuse as well once the patch is run
4	Binned cards stayed on the keeper phone	Sync only added, it never reconciled what had been removed on the other side
5	Zooming in still read the whole page out loud	The capture ignored the zoom. It now crops to what the eye is actually seeing
6	The camera screen had to be scrolled	The screen was taller than a phone, so the read out loud button sat below the fold. It is now sized from the device, and re-measured when the phone is turned

AND SEVEN MORE FROM THE FULL PASS, THE SAME EVENING

#	WHAT WAS WRONG	WHY IT MATTERED
7	The box showing what Recall read never opened	One shadowed name: the element sat in a const called `found`, and inside the try a second `found` took the date from the letter. So the text was never shown, and on a letter with no date that same line threw and the app said it could not read a letter it had read perfectly. Never filing silently is the claim, and it was quietly broken
8	A photo taken on her phone never reached the family	The picture was uploaded and then attached with an update, which the database refuses from a keeper phone on purpose. The card arrived without its picture and the file sat orphaned in storage. The photo now goes up first, so its path travels with the card
9	A page could run this release's script beside modules from the release before	Only the entry scripts carried a version, so the browser was free to keep older modules for ten minutes. This is what made defect 8 look unfixed at first
10	A store that would not open left an empty Today screen with a toast over it	Empty looks exactly like lost cards. It has its own screen now, with both ways out and an honest word about what starting fresh costs

11	Four questions were asked with a browser confirm box	The browser's language, the browser's type size, cannot be read out loud, and it blocks the page while it is up. Both apps ask their own questions now, and the keeper one speaks the question
12	The stripped numbers ate the line break after them	The next sentence was glued onto the marker and read out loud as one run-on sentence
13	Three contrast failures against our own rule	The selected tab at 3.9 to 1, white on the bright orange at 3.1, a summary at 4.39. This app is for people who cannot read small print, so 4.5 is the rule it set itself

05

Requirements, group by group

STATUS AGAINST THE NUMBERED REQUIREMENTS IN DOCS/ANALYSIS.MD

GROUP	WHAT IT COVERS	MUST DONE	REMAINING
R1	Cards, the core of the store	5 of 5	R1.8 spoken questions is on the cut list, deliberately
R2	Camera intake	5 of 5	Accuracy of R2.3 and R2.4 on real paper is still Luke's to sign off
R3	Reminders	2 of 2	R3.5, a notification while the app is closed, is not built and is no longer claimed anywhere in the app
R4	Accounts and sync	1 of 1	R4.5 export is on the cut list
R5	The family interface	6 of 6	R5.8 invites and R5.9 the log are written but unexercised
R6	Household, roles, device linking	4 of 4	R6.6, no role may be changed by claiming a code, now holds in the app and in the database, and the test proves it
R7	Her phone, and the lock on it	2 of 2	R7.4, Face ID, needs a phone with a face to sign off. R7.7, encrypting the cards at rest, is deliberately not in this version: the key cannot be her four numbers alone, or forgetting them loses her memory for good
N1-N11	Accessibility and quality	9 of 11	The Axe pass is done and clean: zero violations on every screen of both apps, against WCAG 2.0 and 2.1 A and AA plus 2.2 AA, after three real contrast failures were fixed. What is left is the half a laptop cannot do, an old Android over mobile data and a screen reader
P1-P14	Data protection	9 of 14	Isolation, redaction, no tracking, the log and who has access all hold and are tested. The consent record and the privacy notice are Wednesday
S1-S6	Safety and the medical line	4 of 6	Nothing is filed silently, nothing is summarised, the medical line is on the first screen and on every card. S5 and S6 are wording and review items

06

How it is built

Stack

- **No build step.** Plain HTML, CSS and ES modules, so what is in the repository is exactly what is served. Nothing can break between the two.
- **GitHub Pages** for hosting, free, HTTPS, deploy on push to `main`.
- **IndexedDB** for everything local, including the photo blobs.
- **Supabase** in Frankfurt: Postgres with row level security, magic link sign-in, anonymous sign-in for the keeper phone, and a private photo bucket reached through signed links.
- **Tesseract.js** for OCR on the device, **Web Speech** for reading out loud. Both free, both offline, neither sends a letter anywhere.

Data model

- `households` and `memberships`, with a role of keeper or helper.
- `records` carry the household, six fields, the spoken sentence and a `deleted_at` for the bin.
- `reminders` carry a repeat and who marked them done.
- `device_links`, `invites`, `activity` and `consents` for setup, audit and lawful basis.
- One `SECURITY DEFINER` function backs every policy, which is what stops the memberships table from checking itself and recursing.
- The keeper phone has an identity without ever having a password: an anonymous session, promoted to keeper of one household by a code the family made.

07

Data protection and safety, where we stand

People will put hospital letters, bank statements and identity cards into this. That makes it a processing of special-category data about a vulnerable person, so the constraints are product decisions rather than a compliance appendix.

- **Local by default.** No account, no upload, no controller relationship at all until a helper deliberately links the phone. Article 25 by default, and it is also our best demo.
- **OCR on the device.** A hospital letter never reaches a third party, which is exactly why cloud OCR was rejected even though it is more accurate.
- **The national number is stripped** before any text is stored, along with IBANs. Belgian letters carry a rijksregisternummer and we have no legal ground to hold one.
- **Deliberately not collected:** location, when the keeper opened the app, how long they looked at anything, and read receipts on cards. Reminder status has exactly three values.
- **Every helper action is logged** where the keeper can read it, in plain sentences, and the log has no update and no delete policy.
- **Never file silently, never summarise.** A date read off a letter is always shown and confirmed, and the app speaks the words that are written rather than a paraphrase.
- **Never promise what it cannot do.** The app used to say it would tell you in time. It now says a reminder waits on the Today screen, and that family should call for anything that really matters.

Still to write, both for the end presentation: a short DPIA, since article 35 is triggered twice over here, and a privacy notice in plain language that the app reads out loud.

08

What is blocked, and on whom

> **Nothing is blocked any more**
>
> `db/patch-001-roles.sql` is applied. It stops `claim_keeper_device` from touching a membership that already exists with another role, adds a trigger that refuses to take a household creator out of the helper role, and repairs a membership that was already demoted. Verified in the database rather than assumed: the trigger is there, both functions carry their refusals, and no creator is left in the wrong role.
>
> The patch had a fault of its own first, which is worth writing down. Its recovery statement changes a role, and the trigger created two statements above it refuses role changes unless the caller is a helper in that household. In the SQL editor there is no caller at all, so the patch would have failed on its own last line. The trigger now trusts a session that carries no JWT claims, which is the SQL editor and a migration and never an API request, not even an anonymous one, and the recovery runs before the trigger exists.

EVERYTHING ELSE THAT IS WAITING ON A PERSON

WAITING ON	WHAT	WHY IT MATTERS
Zacharia	Nothing outstanding on the build. Slides for privacy and the roadmap, and the DPIA	Thursday work. The board stays a project manager tool with one reader, deliberately: the team is shown the progress rather than invited to manage it
Luke	The half of the accessibility pass that needs a real device, and Face ID on a phone with a face on it	Axe on a laptop is a floor, not a substitute. An old Android over mobile data and a screen reader are Thursday work
Mattiece	Four decisions inside the business plan, and the care organisation revenue stream	The plan arrived on Tuesday and every table in it adds up. What is left is where it contradicts itself: year one loses money while hitting its own targets, marketing is two different numbers, and two price tiers need a roadmap feature
Julia	Competitor scan, positioning, the slide deck	The SWOT is in. The positioning sentence is what the jury will quote back

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09

Where the remaining hours go

THIRTEEN HOURS, IN THE ORDER THEY SHOULD BE SPENT

WHEN	WORK	WHY THEN
Tuesday, done	The database patch, the demo script, the phone wizard and the lock, the business plan reviewed	Everything that was blocking somebody else is off the list
Wednesday morning	Luke re-tests on a phone, the spoken consent screen, the four business plan decisions and the positioning	The consent screen is the last thing between us and a defensible upload path
Thursday, morning	Accessibility pass with Axe and Lighthouse, then fix what it finds. The DPIA	Graded, and also the product promise. Nothing here can wait until after the freeze

Thursday, afternoon	Slides, dev freeze at three, one session with a real person over seventy, then the last full run and leaving everything ready	After three the code stops moving and only the words change. Friday morning is the presentation and nothing else, so the phone and the database go into it ready
Friday morning	Present. And write down every question the jury asks	Nothing else is scheduled, on purpose

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Risks worth watching

RISK	WHAT WE DO ABOUT IT
Leftover test data shows up in the demo	Two empty households sit in the project from the afternoon when one press made three. Luke deletes them from the family app, and the stale second keeper phone goes with the wizard
A free Supabase project pauses after about a week idle	Local first means the demo still runs. Waking it is on the Friday checklist
The camera is blocked or absent in the demo room	Choose a photo exists, and a photograph of a letter on the phone is the fallback
OCR misreads a folded letter	Nothing is filed silently. The date is always shown next to the photo and confirmed
Six fixes in one afternoon break something that used to work	The security test runs against the live project, and Luke re-tests the whole flow rather than the six fixes one by one
The jury asks whether it rings when the app is closed	It does not, we say so, and web push with a scheduled sender is on the roadmap
Scope creeps back to three modules	The non-goals list and the roadmap slide, for anyone who asks

11

Appendix: what is in the repository

FILES THAT MATTER

`index.html`	keeper app, six screens
`helper.html`	family app
`js/app.js`	keeper routing, today, camera flow, sync, linking
`js/helper.js`	family app logic, households, the square
`js/store.js`	IndexedDB: cards, photos, reminders
`js/cloud.js`	the only file that talks to Supabase

WHERE THE PROJECT IS WRITTEN DOWN

Requirements analysis	in the repository, so it moves with the code
Two interfaces, one household	the household model, roles, permissions, the surveillance boundary
Notion	sprint board with a timeline, a board by day, a Kanban and a daily log that is written at the end of every session
This report	state of play, what testing found, blockers, the remaining hours

`js/install.js`	the home screen offer, and the iPhone fallback
`js/ocr.js`	OCR, date finding, redaction
`js/camera.js`	camera, magnifier, cropping to the zoom
`js/speech.js`	reading out loud, per language
`db/schema.sql`	tables, functions, policies, storage
`db/patch-001-roles.sql`	the blocker in section 08
`tests/rls_test.py`	twenty five checks against the live project
`docs/analysis.md`	the requirements, R1 to R6, N, P and S

Today, in commits

- Connect the app to the Supabase project in Frankfurt
- Add the security test, all twenty three checks passing
- Fix the keeper phone never seeing the family cards
- Fix three households from one, and a role demotion nobody could see
- Binned cards now disappear from the keeper phone
- Read and keep only what the zoom is showing
- Link the keeper phone with a square, and offer the home screen
- Camera screen fits the device it runs on
- Say what reminders actually do, and test the demotion bug
- Put the family and household requirements in the repository
- State of play, version 2
- Move Recall to another phone, and an optional code on it
- Patch 001 would have been refused by its own trigger
- State of play, versions two and after
- A way out when the local store is wedged
- Recall asks its own questions now
- The box showing what Recall read never opened
- A photo taken on her phone never reached the family
- One version number for every file the browser caches
- The contrast the project promised, everywhere
- The test plan, and what it found

Recall · state of play, version 2 · 8 September 2026 · live at zacharia-vives.github.io/recall